# Pathway Enrichment Analysis of Anti-Fibrotic Therapy in HER2-negative Breast Cancer

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BCB420

# Dataset: GSE255359 Quintela-Fandino et al. 2024

- HER2-negative breast cancer cohort
- RNA-seq (108 tumour samples)
  - 76 baseline
  - 32 after a nintedanib run-in
- **MeCo** = ECM stiffness gene signature
- MeCo score ↓ after treatment by 25% ($P < 0.01$)
- Nintedanib targets PDGFR / FGFR / VEGFR

PRECISION MEDICINE AND IMAGING | NOVEMBER 15 2024

**High Mechanical Conditioning by Tumor Extracellular Matrix Stiffness Is a Predictive Biomarker for Antifibrotic Therapy in HER2-Negative Breast Cancer**

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+ Author & Article Information

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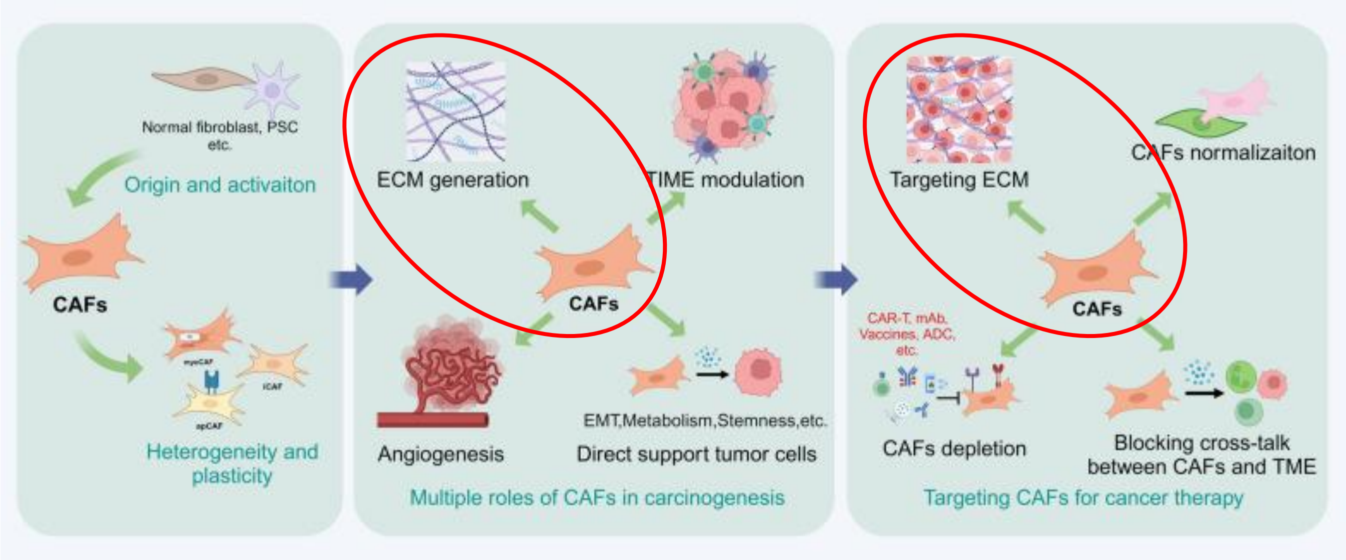
Pre-treatment → Nintedanib run-in → Post-treatment → Chemotherapy

Quintela-Fandino et al. 2024

# Biological Context

- CAFs remodel the extracellular matrix (ECM)
- ECM stiffness activates integrin signalling
- Promotes tumour growth and resistance

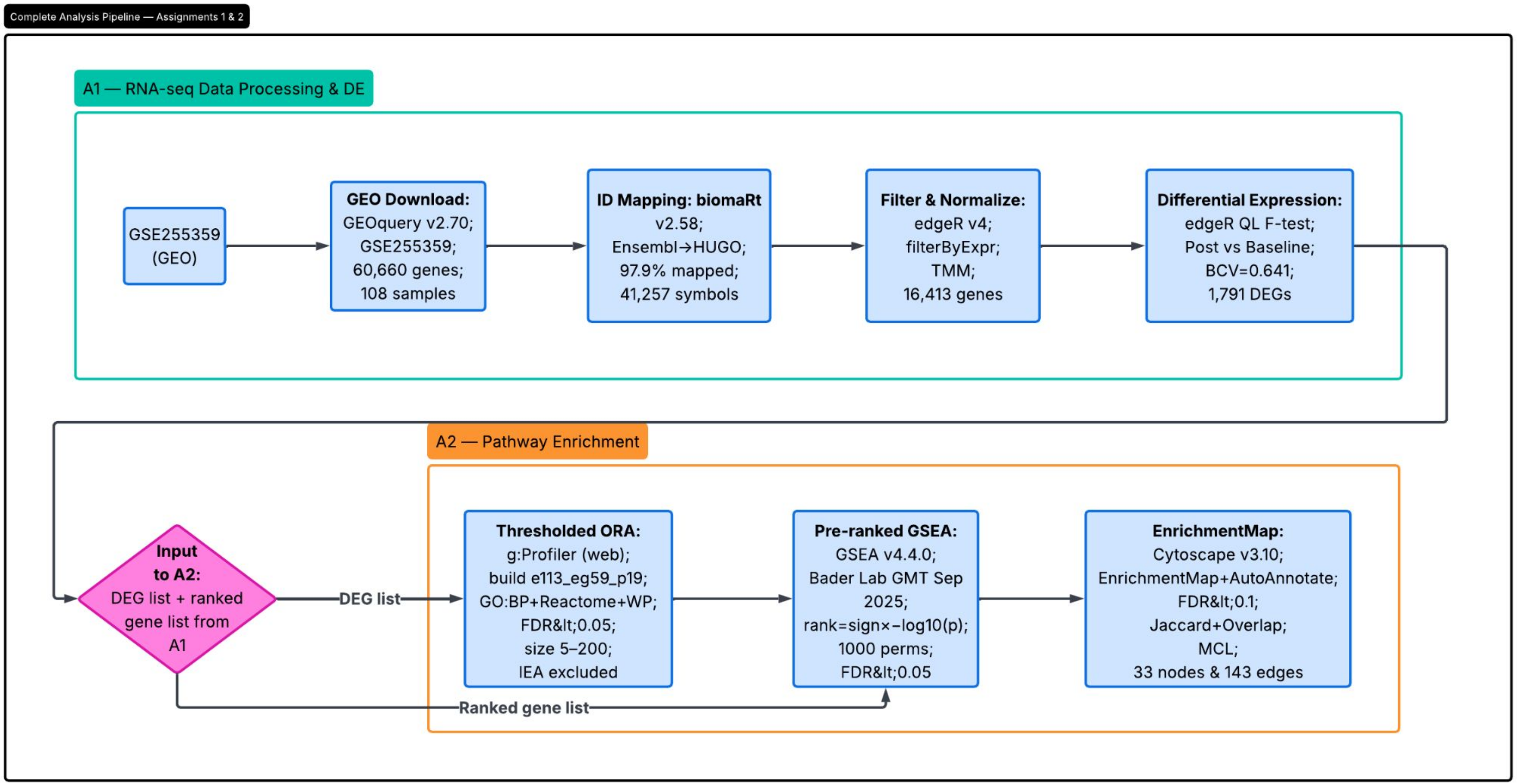



Jia et al. 2025

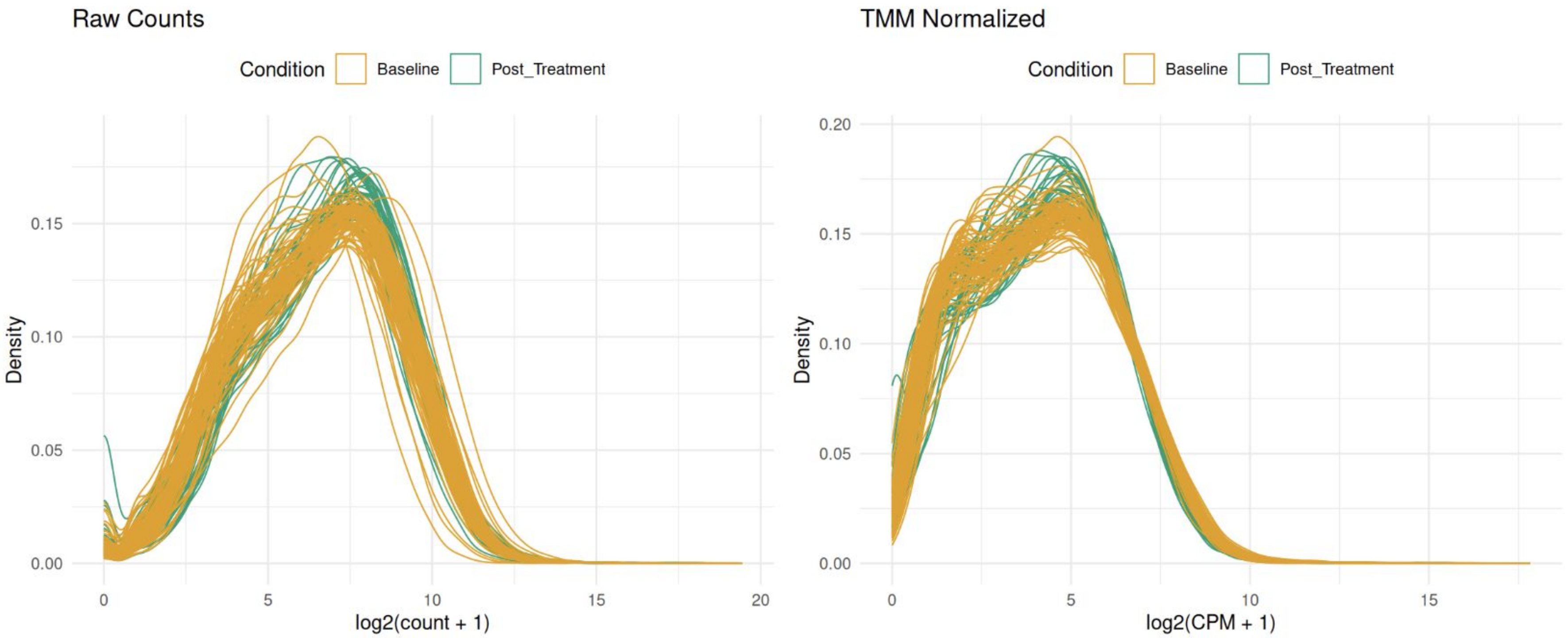
Biological Question:

**What pathways are remodeled by nintedanib, and do they match the expected mechanism of ECM stiffness reversal?**

# Pipeline Overview

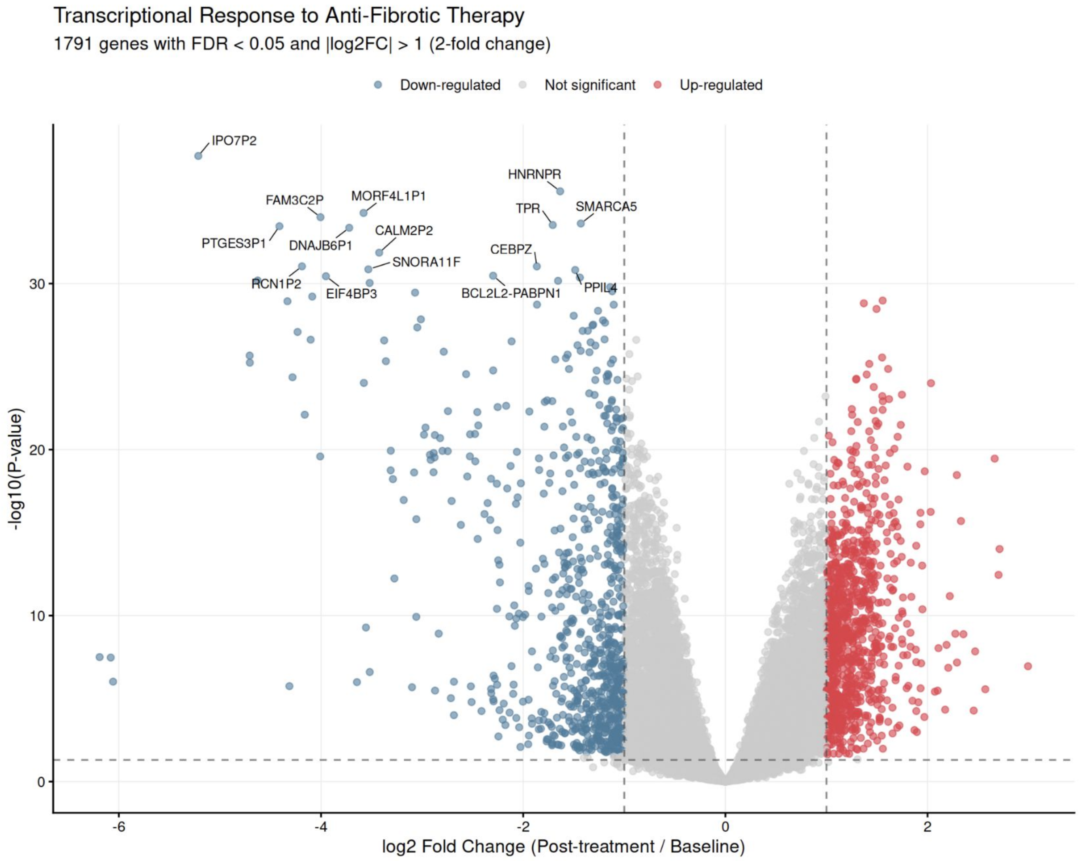


# Normalization



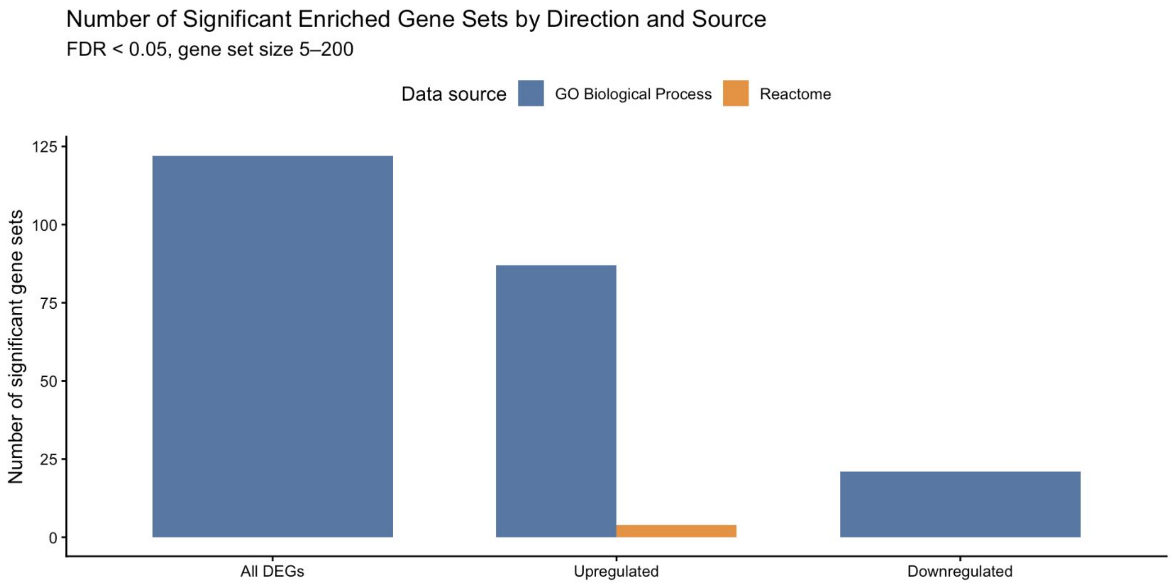
# Differential Expression

- 108 RNA-seq samples
- 1,791 DEGs
    - 973 upregulated
    - 818 downregulated
- FDR < 0.05, |log2FC| > 1

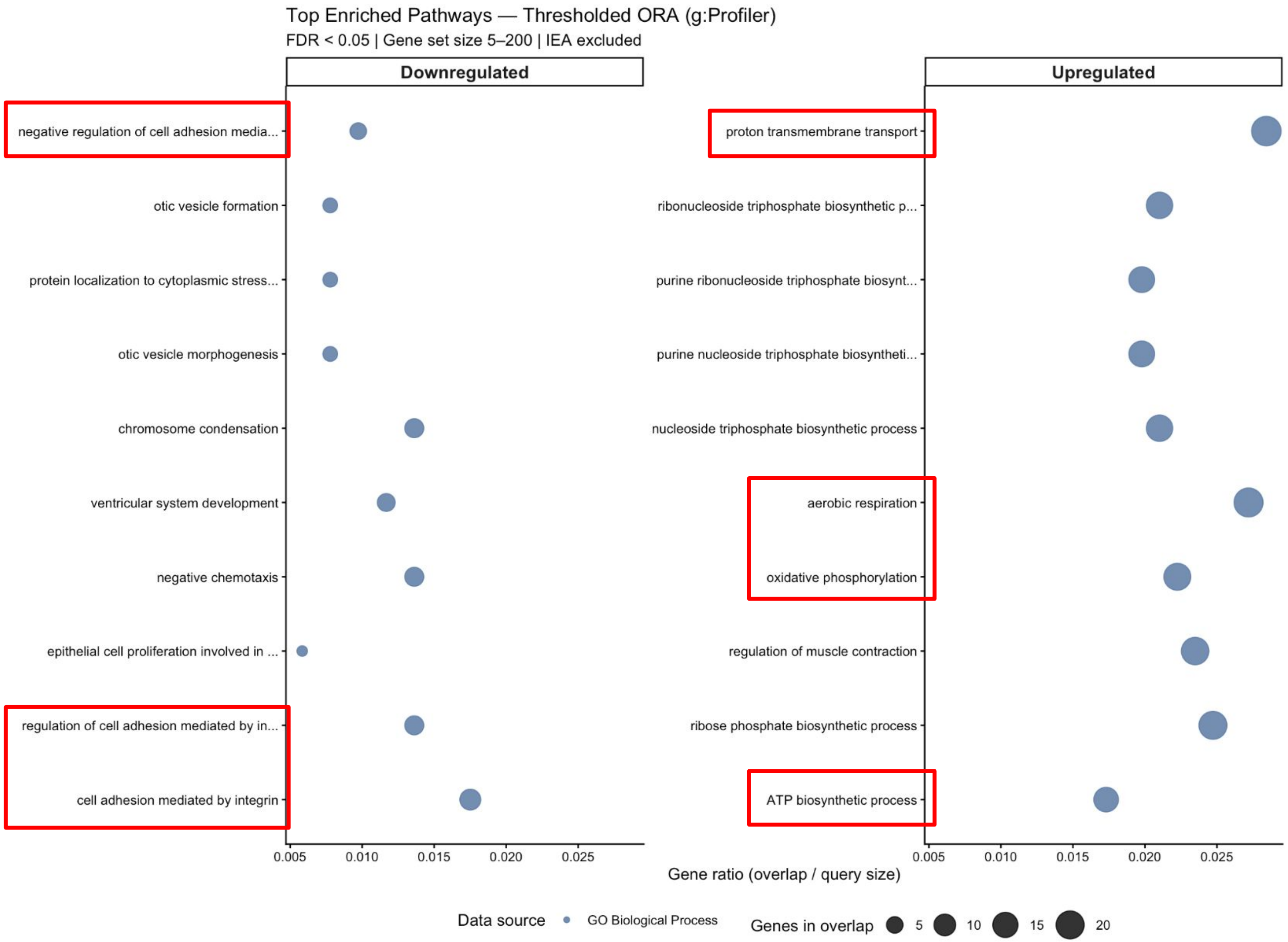


# ORA Results

- Majority of terms from GO:BP
- Minimal contribution from Reactome
- No enrichment from WikiPathways
- Indicates broad, distributed biological signal

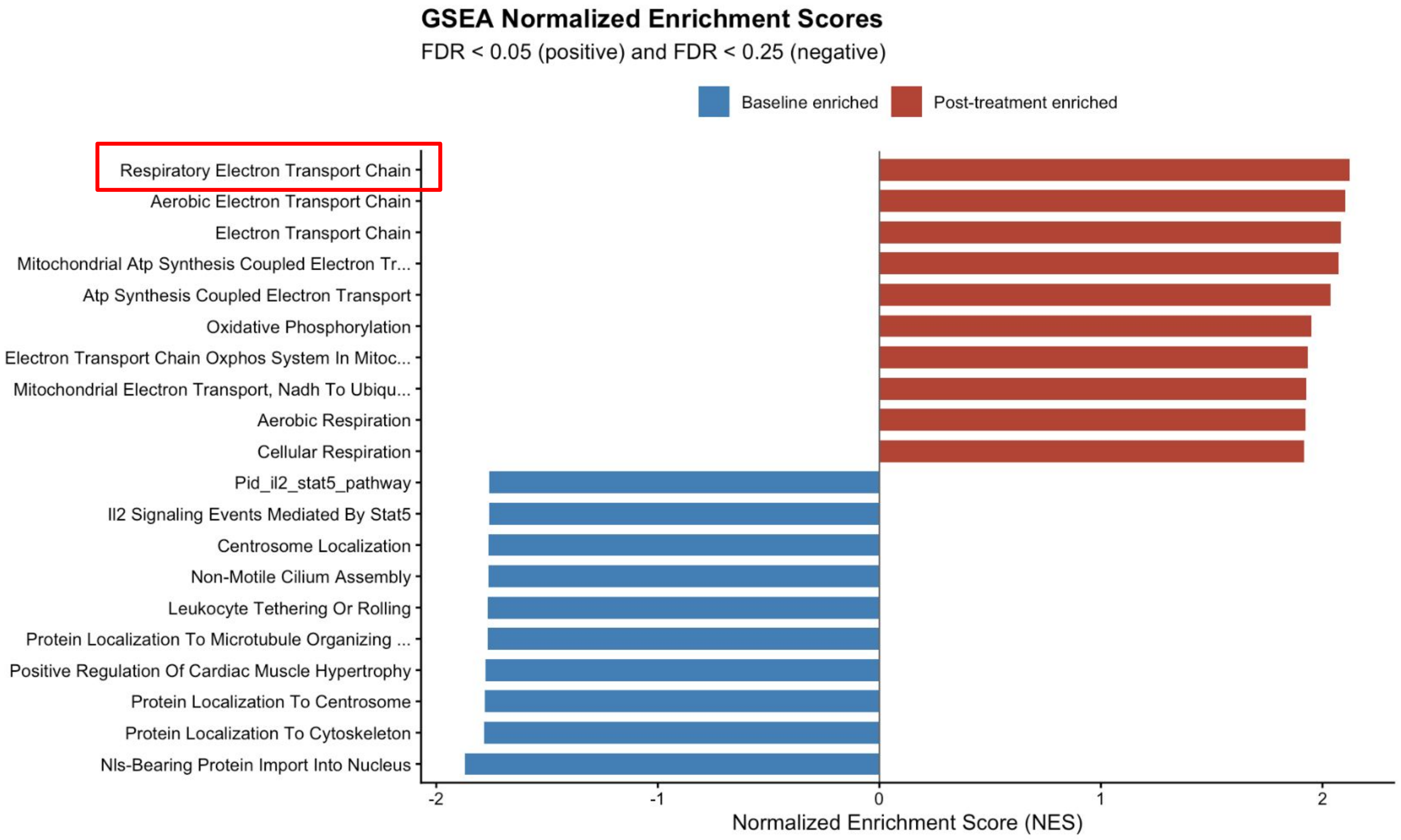


# ORA Results



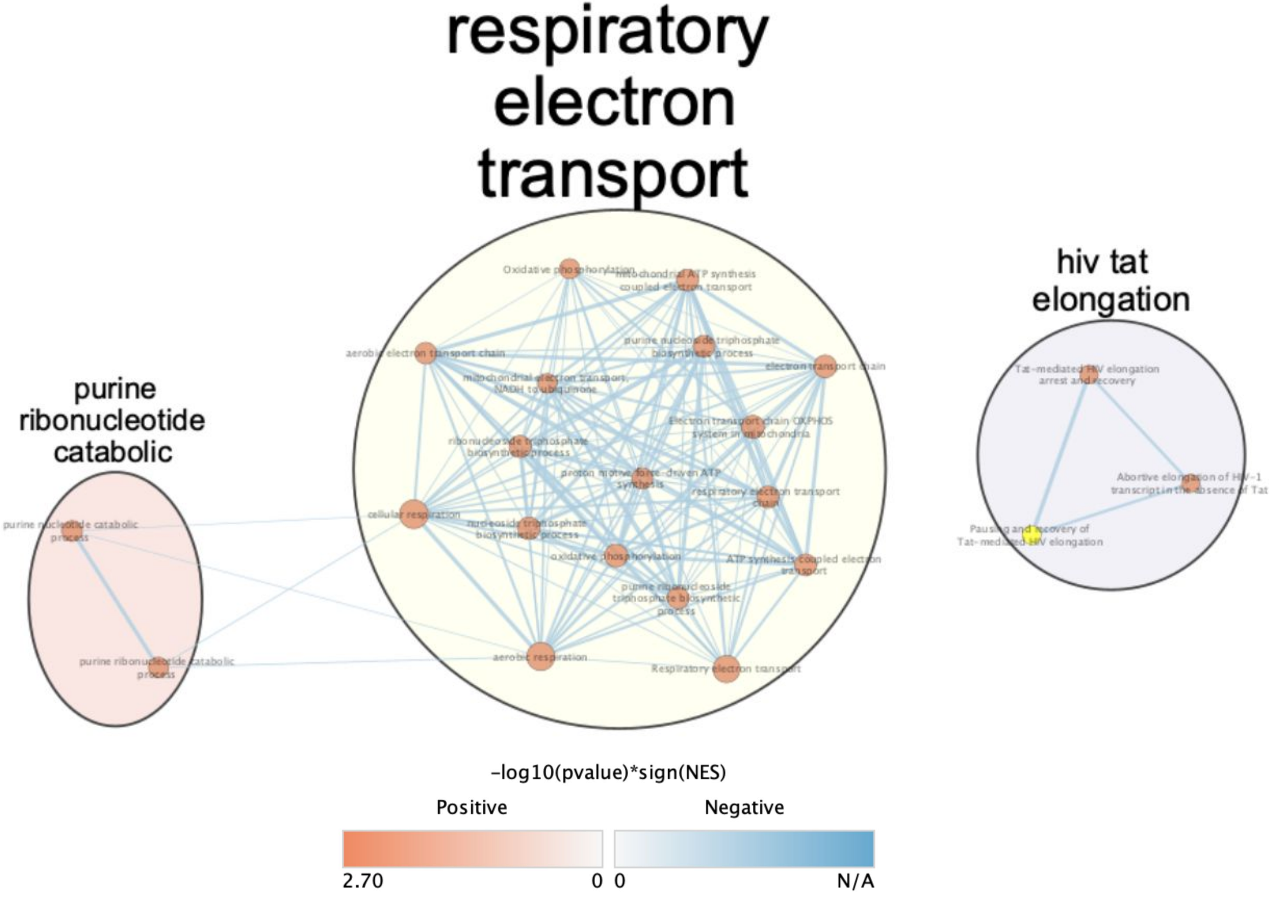
Top Enriched Pathways — Thresholded ORA (g:Profiler)
FDR < 0.05 | Gene set size 5–200 | IEA excluded
Downregulated
negative regulation of cell adhesion media...
otic vesicle formation
protein localization to cytoplasmic stress...
otic vesicle morphogenesis
chromosome condensation
ventricular system development
negative chemotaxis
epithelial cell proliferation involved in ...
regulation of cell adhesion mediated by in...
cell adhesion mediated by integrin
0.005
0.010
0.015
0.020
0.025
Upregulated
proton transmembrane transport
ribonucleoside triphosphate biosynthetic p...
purine ribonucleoside triphosphate biosynt...
purine nucleoside triphosphate biosyntheti...
nucleoside triphosphate biosynthetic process
aerobic respiration
oxidative phosphorylation
regulation of muscle contraction
ribose phosphate biosynthetic process
ATP biosynthetic process
0.005
0.010
0.015
0.020
0.025
Gene ratio (overlap / query size)
Data source
GO Biological Process
Genes in overlap
5
10
15
20

# GSEA Results



# EnrichmentMap

33 nodes | 143 edges



# ORA vs GSEA Comparison

| ORA Results | GSEA Results |
|---|---|
| Oxidative phosphorylation ↑ (top terms) | Oxidative phosphorylation ↑<br>(all significant sets) |
| Integrin adhesion ↓ (21 terms) | Not detected at FDR < 0.05 |
| Misses weak pathways | Detects IL-2/STAT5, estrogen (FDR < 0.25) |

**Similarity:** OXPHOS ↑

**Difference:** discrete vs distributed signals

# Conclusions

**Biological Question:** What pathways are remodeled by nintedanib?

- Oxidative Phosphorylation ↑ (dominant response)
- Integrin adhesion ↓
- Matches ECM stiffness reversal
- Consistent across ORA, GSEA, EnrichmentMap

# References

Jia, H., Chen, X., Zhang, L., & Chen, M. (2025). Cancer associated fibroblasts in cancer development and therapy. Journal of hematology & oncology, 18(1), 36. https://doi.org/10.1186/s13045-025-01688-0

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